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A Deep Learning based Bangladeshi Vehicle Classification using Fine-Tuned Multi-class Vehicle Image Network (MVINet) Model

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***Abstract*—The classification of vehicles is an important task in many applications, including traffic management, surveillance, and law enforcement. In Bangladesh, vehicle classification has become increasingly challenging due to the rapid growth in the number of vehicles on the roads. In this study, a novel deep learning-based Bangladeshi vehicle classification model using fine-tuned Multi-class Vehicle Image Network (MVINet) based on DenseNet201 with four added layers has been proposed. Our approach uses convolutional neural networks (CNNs) to extract features from images of vehicles. To make our dataset, we employ a YOLO (You Only Look Once) model for initial vehicle detection, followed by manual filtering and augmentation of the dataset to improve classification accuracy. The proposed method demonstrates superior performance compared to manual data collection methods. We evaluated our approach using standard metrics such as accuracy, precision, recall, and F1 score. Our model achieved an accuracy of 97.06%, precision of 97.17%, recall of 97.24%, and F1-score of 97.12% indicating high performance in classifying vehicles. We also compared our approach with state-of-the-art techniques and found that the proposed deep learning-based approach outperformed them significantly. The proposed approach has the potential to improve traffic management and law enforcement in Bangladesh. It can be used to monitor traffic flow, detect traffic violations, and identify stolen vehicles. Overall, this study demonstrates the effectiveness of deep learning in Bangladeshi vehicle detection and classification and highlights the importance of using advanced technologies to address the challenges of a rapidly changing transportation landscape.**

***Index Terms*—vehicle detection & classification, transfer learning, densenet201, mvinet, yolo, smart transportation**

I. Introduction

Vehicle classification plays a crucial role in Intelligent Transportation Systems (ITS) whice refer, traffic monitoring, and urban planning [1]. In recent years, deep learning techniques have gained significant attention in computer vision tasks, including image classification [2]. This research focuses on improving vehicle classification accuracy by leveraging transfer learning and advanced data augmentation techniques.

Deep learning (DL)-based vision-based vehicle categorization algorithms have drawn a lot of attention from researchers recently since they provide a practical and flexible solution for meeting the needs of expanding ITS applications. Numerous works have been proposed in this domain that use DL techniques for vision-based vehicle classification.

In [3], the authors propose a two-stage approach to detect and classify vehicles in images. In the first stage, a deep neural network is used to detect the presence of vehicles in an image. The authors used a Convolutional Neural Network (CNN) for this purpose, which was trained on 10 classes of images containing vehicles. Their proposed VGG16 [3] and VGG19 [3] model achieved accuracy of 75.15% and 79.57%, respectively. Another study [4] shows accuracy of 87% using pre-trained DenseNet on their 6 classes of vehicle image dataset. They also showed AlexNet [5] and VGG16 demonstrates lower accuracy of 73% and 70% on their dataset. This article [6] discusses a method for vehicle classification using deep learning, specifically designed for low-quality images that are captured by low-cost cameras or smartphones. Their proposed model showed 92.2% accuracy on their dataset. [7] This research focuses on deep learning-based system for efficient emergency vehicle classification in real-time traffic monitoring using DenseNet121 with 92.2% accuracy. For both vehicle type and colour classification, the authors used a CNN structure as a classifier in this study [8]. Their vehicle classification system takes a single vehicle image as input for classification. TensorFlow's resize feature is used to reduce the original car image to 32x32 pixels. 686 photos make up the training set, while 228 photos make up the testing set. The proposed method's accuracy in classifying colours is 70.09%, while its accuracy in classifying vehicles is 81.62%.

Since all the above mentioned studies work with less classes of vehicle images and gets lower accuracy. Therefore, to address the above mentioned limitations, our model is proposed. The novelty and contributions of the proposed model are:

1) Own dataset made with YOLO and manual annotation combined with publicly available dataset made a huge dataset with 17 distinct classes of vehicles from Bangladesh.
2) Proposed methodology based on transfer learning and fine tuning for Bangladeshi vehicle classification which

can detect seventeen or more classes of vehicles with high accuracy.
3) Comparative analysis with the state-of-the-art techniques

The rest of the article is structured as follows: Section II describes detailed explanation of the research methodology. Section III explains about the proposed model, MVINet. The result analysis of the proposed model is discussed in Section IV. Finally, the paper is concluded in Section V.

II. Research Methodology

The suggested deep learning-based Bangladeshi automobile classification system is described in this section along with the methodology employed for the study, which covers dataset clarification, image preprocessing procedures, and data augmentation approaches. The overall research methodology is summarized in Figure 1.

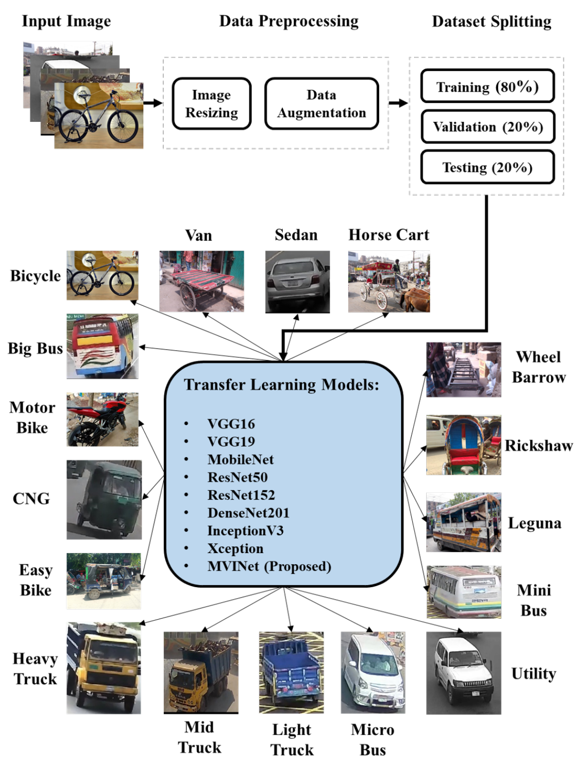

Fig. 1. Diagram of the entire research process

The dataset consists of Bangladeshi vehicle images which are classified into seventeen classes for evaluation of the proposed model. However, the primary focus of the research was to examine the classification performance for the seventeen-class classification. The research was explored with several classification models such as the VGG16 [9], VGG19 [9], DenseNet201 [10], ResNet50 [11], Inception [12] and Xception [13] models as well as our proposed MVINet model.

A. Data Collection and Preparation

To obtain diverse and realistic vehicle data, we employed a YOLO model to detect and extract images of vehicles from street camera videos captured at 30 different locations for a period of 24 hours. The YOLO model offers a real-time object detection system [14], which is computationally efficient and offers high localization accuracy. The video was processed using the YOLO algorithm to detect vehicles in the video frames. The detected vehicles were then cropped and saved into their respective class directories.

Our dataset consists of 10 types of vehicles with 1,000 images in each class. We have also incorporated 7 classes from the publicly available dataset Poribohon-BD [15] with our dataset. The combination of both dataset has made a new huge dataset of 17,000 images of 17 different classes. We collected data from both daytime and nighttime videos to ensure that our dataset is diverse and robust. Also, we have data from diverese weather condition.

The dataset consists of different types of Bangladeshi vehicles which are classified into seventeen classes:

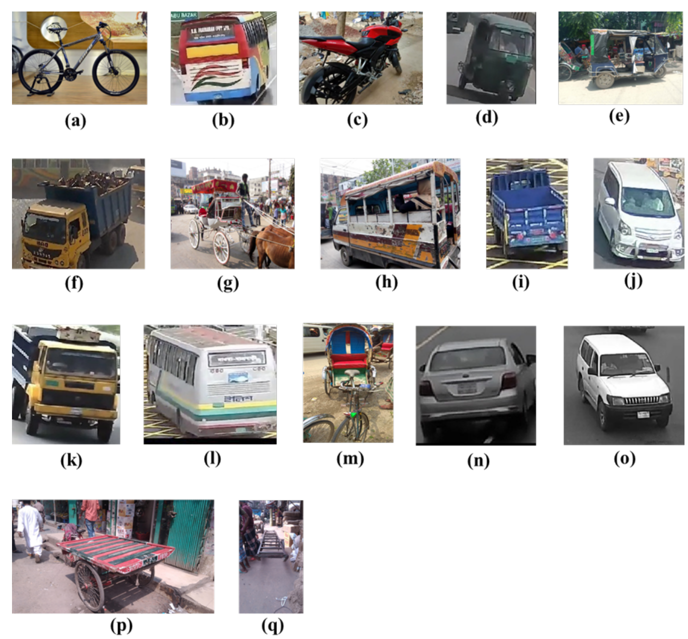

Fig. 2. Samples of 17 classes. (a) Bicycle, (b) Big Bus, (c) Motorbike, (d) CNG, (e) Easy Bike, (f) Heavy Truck, (g) Horse Cart, (h) Leguna, (i) Light Truck, (j) Micro Bus, (k) Mid Truck, (l) Mini Bus, (m) Rickshaw, (n) Sedan, (o) Utility, (p) Van, (q) Wheel Barrow

But only 20% of the total photos were used for testing, while 80% were used for training. In order to avoid overfitting, 20% of the training dataset, which makes up 80% of the dataset, is used for validation. Detailed dataset description is presented in Fig. 3 and some sample images are shown in Fig. 2.

B. Image Pre-processing and Augmentation Procedure

It is essential to correctly prepare the input data before using deep learning models. The quantity and amount of noise in the raw data can fluctuate, which can make it challenging to employ in deep learning models. Additionally, a balanced dataset is crucial for multi-class images because an unbalanced dataset would result in a biased overfitted model. These measures were taken to overcome these issues:

1) Manual Filtering: After the initial data collection, we manually filtered the vehicle images to obtain 10 distinct classes. This manual filtering process was essential for refining the dataset, as it helps to eliminate noisy or irrelevant data, which may negatively impact the classifier's performance.

2) Data Augmentation: We further augmented the dataset using state-of-the-art data augmentation techniques such as rotation, scaling, flipping, and random erasing. These transformations increased the dataset's diversity, enabling the classifier to generalize better and avoid overfitting. Data augmentation is particularly useful when working with limited datasets or when attempting to enhance the model's robustness to various real-world conditions. Below the bar chart shows the comparison between real and augmented data in Fig. 3.

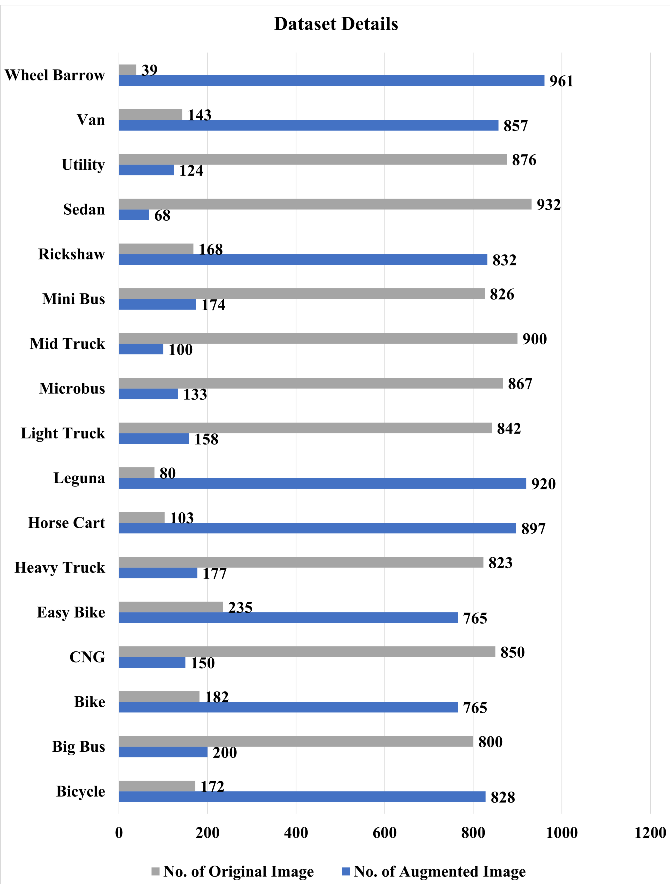

Fig. 3. Details of our dataset

3) Transfer Learning and Model Training: For the classification task, we employed transfer learning using a pre-trained convolutional neural network (CNN) model. Transfer learning enables the reuse of knowledge gained from solving similar tasks, thereby reducing the required training time and computational resources. We fine-tuned the pre-trained model on our cropped and augmented dataset, adjusting its weights to optimize its performance for the vehicle classification task.

4) Advantages of Transfer Learning over Manual Data Collection: Transfer learning offers several advantages over manual data collection methods:

- Efficiency: Transfer learning significantly reduces the training time and computational resources required to achieve high classification accuracy, as the model has already learned relevant features from similar tasks.
- Robustness: Leveraging a pre-trained model improves the classifier's robustness to various real-world conditions and reduces the likelihood of overfitting.
- Scalability: Transfer learning allows for easy adaptation to new classes or variations in the data, making it a flexible and scalable approach to image classification tasks.

III. Analysis of the Proposed Multi-class Vehicle Image Network (MVINet) Model

In this study, we propose a novel deep learning-based approach for Bangladeshi vehicle classification using a fine-tuned and modified DenseNet201 model. DenseNet is a type of convolutional neural network (CNN) that has shown excellent performance in image classification tasks. We chose DenseNet201 because it is a powerful pre-trained model that has been used successfully in various image classification tasks.

To further improve the classification accuracy, we have added a new convulational layer, a batch normalization layer, an activation layer and a global average pooling layer after the last convolutional block of the model shown in Fig. 4, and Fig. 5.

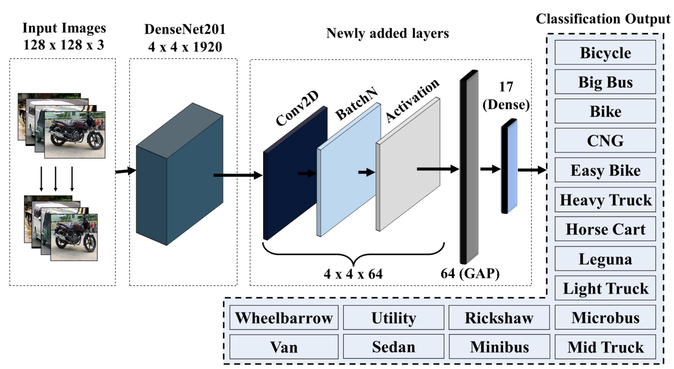

Fig. 4. The proposed model architecture

conv2d_102 (Conv2D)	(None, 4, 4, 64)	122944	relu[0][0]
batch_normalization_102 (BatchN	(None, 4, 4, 64)	256	conv2d_102[0][0]
activation_95 (Activation)	(None, 4, 4, 64)	0	batch_normalization_102[0][0]
global_average_pooling2d_5 (Glo	(None, 64)	0	activation_95[0][0]
dense_6 (Dense)	(None, 17)	1105	global_average_pooling2d_5[0][0]

Fig. 5. Newly added layers

Benefit of adding these layers are given below:

1) Convulational Layer: By bringing non-linearity to the network, lowering the number of parameters, learning to identify pertinent characteristics from the input data, and being translation invariant, convolutional layers can increase accuracy. Convolutional layers are able to learn sophisticated and abstract feature representations of the input data by mixing these features, which can increase accuracy on a variety of tasks.

2) Batch Normalization Layer: By enhancing convergence, speeding up learning, serving as a kind of regularisation, and enhancing gradient flow through the network, batch normalisation can increase accuracy. Batch normalisation can help to enhance the generalisation of the network and result in higher performance on a variety of tasks by stabilising the training process and preventing overfitting.

3) Activation Layer: The main advantage of activation layers, a crucial part of neural networks, is that they add non-linearity to the network's output. They aid in identifying intricate patterns in the input data and enhancing the network's expressiveness, improving prediction task accuracy. In deeper networks, where this can be an issue, activation layers also stop the gradient from vanishing during backpropagation. By reducing overfitting and boosting the network's generalisation ability, they also aid in regularising the network.

4) Global Average Pooling (GAP): Global average pooling is an effective technique that can help reduce the number of parameters in the model and prevent overfitting. It works by taking the average of the feature maps of each channel in the last convolutional layer, resulting in a one-dimensional vector that can be fed into the dense layer for classification.

5) Output layer: In addition to global average pooling, we also added a dense layer with softmax activation function. The dense layer is responsible for performing the final classification of the vehicle image. The softmax activation function ensures that the output of the dense layer represents a valid probability distribution over the possible classes of vehicles.

We evaluated the performance of our proposed model using standard metrics such as accuracy, precision, recall, and F1 score. Our model achieved an accuracy of 97.06%, precision of 97.17%, recall of 97.24%, and F1-score of 97.12%, indicating high performance in classifying vehicles. The results of our experiments demonstrated that the proposed model with global average pooling and dense layer performed significantly better than the baseline model without these modifications.

IV. Result Analysis and Discussion

A. Model's Performance Assessment Matrices

The proposed MVINet model may be evaluated using four evaluation criteria : : (i) Accuracy (A), (ii) Recall (R), (iii) Precision (P), and (iv) F1-score (Fs) to see how well it performs classification. These assessment metrics are computed using following formulas:

$$A = \frac{N_{\text{TP}} + N_{\text{TN}}}{N_{\text{TP}} + N_{\text{FN}} + N_{\text{FP}} + N_{\text{TN}}} \quad (1)$$

$$R = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \quad (2)$$

$$P = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \quad (3)$$

$$F_{\text{s}} = \frac{2 * N_{\text{TP}}}{2 * N_{\text{TP}} + N_{\text{FN}} + N_{\text{FP}}} \quad (4)$$

where N_{TP} denotes the number of correctly classified vehicle images, N_{TN} represents when the classifier correctly predicts a negative class (i.e., the actual class is negative and the classifier predicts negative), N_{FP} denotes the number of misclassified vehicle images, and N_{FN} refers to the number of instances in which the classifier incorrectly predicts a positive class (i.e., the actual class is positive but the classifier predicts negative).

B. Experimental Analysis

In this research, the Tensorflow and Keras framework with Python version 3.6 was utilized, which is running on the Anaconda distribution platform. Table I presents the hyper-parameters of the suggested model. The Hyperparameters are the values, that are set before training the model. These parameters significantly impact the performance of the model. Therefore, it is a crucial task to choose appropriate values for the hyperparameters of the model.

TABLE I
HYPERPARAMETERS OF THE PROPOSED MODEL

Hyperparameters	**Values**
Optimizer	Adam
Learning-rate	0.0001
Batch-size	16
Epochs	30
Loss Function	Categorical Cross Entropy
Activation	ReLU
Image Size	128 x 128 x 3

In the model, the Adam [16] optimization algorithm is utilized. The Adam optimizer incorporates some ideas and techniques from Adaptive Gradient Algorithm (AdaGrad) [17] and Root Mean Square Propagation (RMSProp) [18]. It can efficiently optimize the non-convex objective functions. The learning rate is set to 0.0001, which helps converge the model. The learning rate determines the step size in the parameter space during gradient descent, and it is a crucial hyper-parameter to tune. The optimal batch is set to 16. Batch size is a hyper-parameter that affects the convergence speed and the memory requirements during training. The model was trained for 30 epochs. The number of epochs is a hyper-parameter that controls how many times the model will see the data during training. In addition, the Categorical Cross-Entropy loss function is used for multi-class classification purposes. Finally, the ReLU is utilized as an activation function for the model that provides better performance. However, the outperforms of the model was better due to by using tuned hyper-parameters [19].

C. Classification Performance of the Proposed Model

In Table II the classification report of the Multi-class Vehicle Image Network (MVINet) Model shows the performance metrics for each class of vehicles in terms of precision, recall, F1-score, and support.

The model seems to perform well for most classes of vehicles. The precision values for all classes are relatively high, ranging from 0.89 to 1.00, indicating that the model is accurate

TABLE II
CLASSIFICATION PERFORMANCE OF THE PROPOSED MODEL

Classes	Precision	Recall	F1-score	Support
Bicycle	1.00	1.00	1.00	188
Big Bus	0.92	0.96	0.94	195
Bike	0.99	1.00	1.00	198
CNG	0.93	0.97	0.95	215
Easy Bike	0.99	0.99	0.99	207
Heavy Truck	1.00	0.96	0.98	186
Horse Cart	1.00	1.00	1.00	206
Light Truck	0.98	1.00	0.99	180
Microbus	0.97	0.99	0.98	221
Leguna	0.99	0.97	0.97	200
Mid Truck	0.98	0.91	0.94	195
Mini Bus	0.95	0.94	0.93	199
Rickshaw	0.89	0.98	0.95	202
Sedan	0.97	0.91	0.93	212
Utility	1.00	1.00	1.00	189
Van	1.00	1.00	1.00	202
Wheel Barrow	0.96	0.95	0.96	205

in predicting the correct class for the majority of instances. The recall values are also generally high, ranging from 0.91 to 1.00, indicating that the model is effective in identifying the majority of instances for each class. The F1-scores are also high for most classes, ranging from 0.93 to 1.00, which indicates that the model is performing well in terms of both precision and recall for most classes of vehicles. The support values for each class vary but are generally consistent, ranging from 180 to 221 instances. Overall, the classification report suggests that the MVINet model is performing well in accurately classifying most classes of vehicles, with high precision, recall, and F1-scores for the majority of classes.

Fig 6 shows the loss and accuracy curve of the training, validation, and testing set of our proposed model MVINet on our dataset. Acquiring more training data will help the model generalize better by exposing it to a wider range of examples and reducing the likelihood of overfitting.

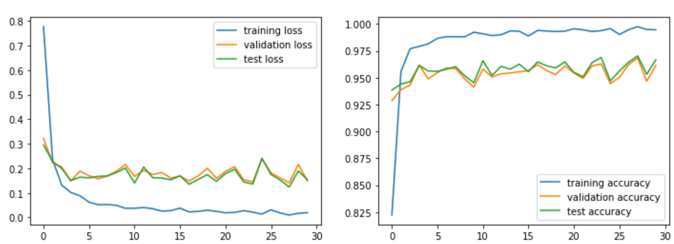

Fig. 6. Loss and accuracy curve of the proposed MVINet model

The confusion matrix of the MVINet model is shown in Fig. 7. The categorization outcomes of the test dataset by the model are displayed in the confusion matrix. The confusion matrix demonstrates that the majority of the classes, which are displayed on the diagonal of the matrix, are correctly categorised.

In addition, the proposed model and six others (VGG16, VGG19, InceptionV3, DenseNet201, ResNet50, Xception) classification models were evaluated by utilizing our dataset

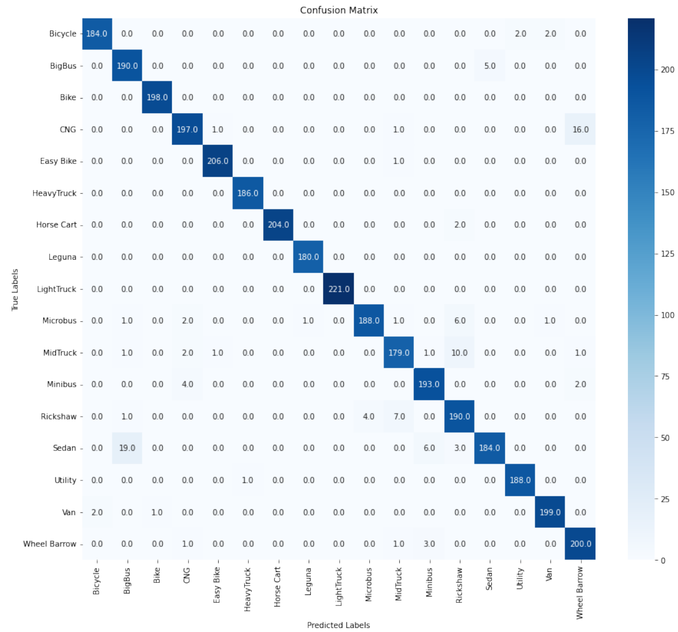

Fig. 7. Confusion matrix of the proposed model

to ensure the validity of the proposed model.

TABLE III
CLASSIFICATION REPORT OF DIFFERENT MODELS ON OUR DATASET

Models	Accuracy(%)	Precision	Recall	F1-score
VGG16	72.6	0.721	0.735	0.713
VGG19	77.6	0.781	0.785	0.773
MobileNet	91.4	0.916	0.915	0.915
InceptionV3	92.4	0.925	0.926	0.924
ResNet50	93.0	0.932	0.935	0.933
ResNet152	93.6	0.934	0.935	0.934
Xception	94.6	0.945	0.949	0.967
Densenet201	95.2	0.955	0.951	0.959
Proposed Model	**97.06**	**0.972**	**0.972**	**0.971**

Table III presents the classification performance of the models. It is observed from Table III, Xception, ResNet152 and ResNet50 models achieved moderate to high accuracy (94.6%, 93.6% & 93.0%) values respectively. On the other hand, the DenseNet201 model attained high accuracy of 95.2% than other models. DenseNet201 is a deeper variant of DenseNet with 201 layers. Deeper models often have more capacity to learn complex features and patterns, which can be beneficial for transfer learning tasks. By pretraining DenseNet201 on large-scale datasets like ImageNet, the model can capture a rich set of generic visual features that can be fine-tuned or transferred to specific tasks with smaller datasets. So, proposed model MVINet was created based on DenseNet201. Highest accuracy of the proposed model 97.06% indicates that, the model can accurately classify the different types of Bangladeshi Vehicles.

Based on the comparison Table IV of the proposed model's performance with several state-of-the-art techniques for different Bangladeshi vehicle classification provided, it appears that the proposed model MVINet outperforms state-of-the-art techniques in terms of accuracy and F1 score for the dataset it was trained on. Proposed dataset is a unique combination of street camera videos and PoribahanBD dataset. Specifically, the proposed CNN model [6] achieved an accuracy of 0.929 for a surveillance camera dataset, which is higher than the accuracy achieved by the DenseNet model for an HD camera dataset in reference [4]. Additionally, when compared to YOLOV5 in reference [3], the proposed model achieved a higher F1 score of 0.971 for a dataset that combined Our dataset and PoribahanBD, while YOLOV5 achieved an F1 score of 0.790 for the PoribahanBD dataset.

TABLE IV
PROPOSED MODEL IN COMPARISON WITH STATE-OF-THE-ART TECHNIQUES

Ref.	Dataset	Method	No. of Classes	Acc.	F1 Score
[8]	Surveillance video	CNN	4	0.816	—
[3]	PoribahanBD	YOLOV5	10	0.832	0.790
[4]	HD Camera, Kaggle,IndiaAI	DenseNet	6	0.870	—
[6]	Surveillance camera	Proposed CNN model	6	0.929	—
[7]	Analytics Vidhya Emergency Vehicle dataset	DenseNet121	2	0.951	0.938
Proposed Model	**Our dataset+ PoribahanBD**	**MVINet**	**17**	**0.970**	**0.971**

However, it is concluded that the proposed model performed superior classification performance than the state-of-the-art approaches for Bangladeshi Vehicle Classification. The improvement in results is mainly observed due to adding new layers and choosing best hyperparameters for the proposed model.

V. CONCLUSION

After conducting a comprehensive study on Bangladeshi vehicle classification using deep learning techniques, we propose a novel Fine Tuned Multi-class Vehicle Image Network (MVINet) model that achieves superior performance compared to state-of-the-art methods. Our approach employs a YOLO model for initial vehicle detection and manual filtering and augmentation of the dataset to improve classification accuracy. The model achieved an accuracy of 97.06%. Our study highlights the importance of using advanced technologies to address the challenges of a rapidly changing transportation landscape in Bangladesh. The proposed approach can be used for traffic management, surveillance, law enforcement, and vehicle theft prevention. Our research provides insights into the effectiveness of deep learning in Bangladeshi vehicle detection and classification and serves as a foundation for further research in this area. Future work can focus on improving the accuracy of classification for rare classes of vehicles, exploring the effectiveness of transfer learning with other datasets, and integrating the proposed model with real-time traffic management systems for practical applications.

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